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SUMMARY

Product manager for grid-scale BESS with a utility and construction background, eight years at LG Energy Solution moving from systems engineering to product ownership. Originated the 2-hour containerized product that became LG's flagship for AI data center power, with multi-GWh bookings. Works across power systems engineering, techno-economic modeling, and Python tooling to turn site-level constraints into deployable hardware and pricing.

CORE COMPETENCIES

Power Systems & Hardware: Grid-scale and containerized BESS, PCS selection and configuration, UL 9540, black start and islanding, voltage fault ride-through, non-firm interconnection and curtailment, demand response, central plant and data center cooling systems

Modeling & Analysis: Cell-to-system cost modeling, EPC and logistics cost modeling, battery degradation modeling (calendar and cycle aging), system sizing and augmentation planning, Python, Excel

Commercial & Delivery: Product roadmap and business case, pricing strategy, RFP/RFI technical response, vendor and contract manufacturer selection, change order negotiation, utility incentive structuring

EXPERIENCE

LG Energy Solution

Feb 2018 – Present

Senior Product Manager, Grid-Scale BESS

Feb 2021 – Present

- **Expanded product portfolio from 4-hour to 2-hour duration** with an AC-integrated containerized BESS to capture emerging AI data center demand; secured executive and commercial buy-in to establish the platform as LG's flagship offering, driving multi-GWh forward bookings for 2027 delivery.
- **Built bottom-up cell-to-system cost, EPC installation, and logistics models** to set competitive system pricing, evaluate PCS configurations, and quantify the margin, energy density, and transport advantages of the 2-hour architecture.
- **Translated hyperscale and AI data center power constraints** into core BESS requirements: authored specifications for compute step-load transient smoothing, voltage fault ride-through with active recovery, gas turbine pairing, black start and islanding, and capacity management for non-firm interconnection with curtailment.
- **Led PCS vendor selection and owned the UL 9540 listing path;** served as the product interface between engineering, sales, and project execution, aligning volume commitments, cost-down trajectory, development/certification schedules, technical documentation, and commissioning procedures.

Senior Systems Engineer, Grid-Scale BESS (LG Chem)

Feb 2018 – Feb 2021

- **Led product engineering for LG's first prefabricated containerized BESS** (2.86 MWh/unit); partnered with contract manufacturer Jabil from architecture through functional prototype, managing cost, safety, and size/weight tradeoffs to replace bespoke field assembly and deliver nearly 1 GWh across CAISO and ERCOT.
- **Developed Python and spreadsheet degradation models** simulating calendar and cycle aging, C-rate, temperature, and depth of discharge; optimized battery-to-inverter ratios and augmentation schedules to minimize lifetime cost while satisfying developer warranty terms.
- **Directed technical sales engineering for integrators, EPCs, and IPP/developers;** led requirements capture, system sizing, and RFP/RFI technical defense, translating site duty-cycle constraints into pre-construction proposals.

Design-Build Engineering Manager, HVAC

- **Managed end-to-end commercial HVAC delivery on active construction sites:** scope definition, equipment selection, and mechanical and electrical commissioning and startup, coordinating with general contractors, architects, and developers.
- **Owned estimating, procurement, and field operations;** produced mechanical bids, directed trade installer crews, and negotiated in-flight change orders to protect margin and schedule.

Pacific Gas and Electric Company (PG&E)

Dec 2013 – Jul 2016

Energy Solutions Engineer

- **Delivered 2.5 MW of peak demand reduction and 7 GWh of cumulative energy savings** across commercial and industrial portfolios through customer-facing technical audits, incentive structuring, and capital project evaluation.
- **Engineered efficiency and demand-response projects for data centers and critical facilities:** developed technical scopes and savings models for data center cooling retrofits (hot/cold aisle containment, economizers, VFDs), central plant systems (chillers, boilers, AHUs), and behind-the-meter thermal and battery peak-shifting assets.

SELECTED TECHNICAL PROJECT

Institutional Equity & Macroeconomic Research Engine

Python, Gemini API, PyMuPDF, SQLite, Discord API, Railway

Automated synthesis and cited RAG bot converting institutional research into a daily brief and Q&A for 100+ paid subscribers.

- **Engineered an automated pipeline ingesting 100+ research PDFs daily** (16,000+ total); parsed unstructured multi-asset commentary into structured JSON schemas with verbatim anchor quotes, programmatically verifying every extracted metric against source text before database commit.
- **Built the automated “Morning Pulse” briefing, published every trading day** (110+ issues) with a lead theme, desk-call trade board, and economic and earnings calendar drawn from live options chains and agency APIs; a redesigned synthesis pipeline in shadow pilot raised extraction fidelity against source PDFs from 33–60% to 87–100% and cut grouping fragmentation from 30–60% to 9–16%.
- **Built an interactive Q&A agent executing cited answers over the research corpus** (1,470+ production queries); regex-based tool router restricts model actions per query, and a payload provenance guard blocks any figure absent from tool output.

EDUCATION & CREDENTIALS

UC Davis Graduate School of Management

2022

MBA (STEM-designated), Product Management & Strategy

UC Davis

2013

BS Mechanical Engineering, Dean's Honor List

Professional Engineer, California

2016

License #M38303, Thermal and Fluid Systems (NCEES)